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Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles

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Abstract

Deep neural networks (NNs) are powerful black box predictors that have recently achieved impressive performance on a wide spectrum of tasks. Quantifying predictive uncertainty in NNs is a challenging and yet unsolved problem. Bayesian NNs, which learn a distribution over weights, are currently the state-of-the-art for estimating predictive uncertainty; however these require significant modifications to the training procedure and are computationally expensive compared to standard (non-Bayesian) NNs. We propose an alternative to Bayesian NNs that is simple to implement, readily parallelizable, requires very little hyperparameter tuning, and yields high quality predictive uncertainty estimates. Through a series of experiments on classification and regression benchmarks, we demonstrate that our method produces well-calibrated uncertainty estimates which are as good or better than approximate Bayesian NNs. To assess robustness to dataset shift, we evaluate the predictive uncertainty on test examples from known and unknown distributions, and show that our method is able to express higher uncertainty on out-of-distribution examples. We demonstrate the scalability of our method by evaluating predictive uncertainty estimates on ImageNet.

1 Introduction

Deep neural networks (NNs) have achieved state-of-the-art performance on a wide variety of machine learning tasks [35] and are becoming increasingly popular in domains such as computer vision [32], speech recognition [25], natural language processing [42], and bioinformatics [2, 61]. Despite impressive accuracies in supervised learning benchmarks, NNs are poor at quantifying predictive uncertainty, and tend to produce overconfident predictions. Overconfident incorrect predictions can be harmful or offensive [3], hence proper uncertainty quantification is crucial for practical applications.

Evaluating the quality of predictive uncertainties is challenging as the 'ground truth' uncertainty estimates are usually not available. In this work, we shall focus upon two evaluation measures that are motivated by practical applications of NNs. Firstly, we shall examine *calibration* [12, 13], a frequentist notion of uncertainty which measures the discrepancy between subjective forecasts and (empirical) long-run frequencies. The quality of calibration can be measured by *proper scoring rules* [17] such as log predictive probabilities and the Brier score [9]. Note that calibration is an orthogonal concern to accuracy: a network's predictions may be accurate and yet miscalibrated, and vice versa. The second notion of quality of predictive uncertainty we consider concerns generalization of the predictive uncertainty to domain shift (also referred to as *out-of-distribution* examples [23]), that is, measuring if the network *knows what it knows*. For example, if a network trained on one dataset is evaluated on a completely different dataset, then the network should output high predictive uncertainty as inputs from a different dataset would be far away from the training data. Well-calibrated predictions that are robust to model misspecification and dataset shift have a number of important practical uses (e.g., weather forecasting, medical diagnosis).

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There has been a lot of recent interest in adapting NNs to encompass uncertainty and probabilistic methods. The majority of this work revolves around a Bayesian formalism [4], whereby a prior distribution is specified upon the parameters of a NN and then, given the training data, the posterior distribution over the parameters is computed, which is used to quantify predictive uncertainty. Since exact Bayesian inference is computationally intractable for NNs, a variety of approximations have been developed including Laplace approximation [40], Markov chain Monte Carlo (MCMC) methods [46], as well as recent work on variational Bayesian methods [6, 19, 39], assumed density filtering [24], expectation propagation [21, 38] and stochastic gradient MCMC variants such as Langevin diffusion methods [30, 59] and Hamiltonian methods [53]. The quality of predictive uncertainty obtained using Bayesian NNs crucially depends on (i) the degree of approximation due to computational constraints and (ii) *if* the prior distribution is 'correct', as priors of convenience can lead to unreasonable predictive uncertainties [50]. In practice, Bayesian NNs are often harder to implement and computationally slower to train compared to non-Bayesian NNs, which raises the need for a 'general purpose solution' that can deliver high-quality uncertainty estimates and yet requires only minor modifications to the standard training pipeline.

Recently, Gal and Ghahramani [15] proposed using *Monte Carlo dropout* (MC-dropout) to estimate predictive uncertainty by using *Dropout* [54] at test time. There has been work on approximate Bayesian interpretation [15, 29, 41] of dropout. MC-dropout is relatively simple to implement leading to its popularity in practice. Interestingly, dropout may also be interpreted as *ensemble model combination* [54] where the predictions are averaged over an ensemble of NNs (with parameter sharing). The ensemble interpretation seems more plausible particularly in the scenario where the dropout rates are not tuned based on the training data, since any sensible approximation to the true Bayesian posterior distribution has to depend on the training data. This interpretation motivates the investigation of ensembles as an alternative solution for estimating predictive uncertainty.

It has long been observed that ensembles of models improve predictive performance (see [14] for a review). However it is not obvious when and why an ensemble of NNs can be expected to produce good uncertainty estimates. Bayesian model averaging (BMA) assumes that the true model lies within the hypothesis class of the prior, and performs *soft model selection* to find the single best model within the hypothesis class [43]. In contrast, ensembles perform *model combination*, i.e. they combine the models to obtain a more powerful model; ensembles can be expected to be better when the true model does not lie within the hypothesis class. We refer to [11, 43] and [34, §2.5] for related discussions. It is important to note that even exact BMA is not guaranteed be robust to mis-specification with respect to domain shift.

Summary of contributions: Our contribution in this paper is two fold. First, we describe a simple and scalable method for estimating predictive uncertainty estimates from NNs. We argue for training probabilistic NNs (that model predictive distributions) using a proper scoring rule as the training criteria. We additionally investigate the effect of two modifications to the training pipeline, namely (i) *ensembles* and (ii) *adversarial training* [18] and describe how they can produce smooth predictive estimates. Secondly, we propose a series of tasks for evaluating the quality of the predictive uncertainty estimates, in terms of calibration and generalization to unknown classes in supervised learning problems. We show that our method significantly outperforms (or matches) MC-dropout. These tasks, along with our simple yet strong baseline, serve as an useful benchmark for comparing predictive uncertainty estimates obtained using different Bayesian/non-Bayesian/hybrid methods.

Novelty and Significance: Ensembles of NNs, or *deep ensembles* for short, have been successfully used to boost predictive performance (e.g. classification accuracy in ImageNet or Kaggle contests) and adversarial training has been used to improve robustness to adversarial examples. However, to the best of our knowledge, ours is the first work to investigate their usefulness for predictive uncertainty estimation and compare their performance to current state-of-the-art approximate Bayesian methods on a series of classification and regression benchmark datasets. Compared to Bayesian NNs (e.g. variational inference or MCMC methods), our method is much simpler to implement, requires surprisingly few modifications to standard NNs, and well suited for distributed computation, thereby making it attractive for large-scale deep learning applications. To demonstrate scalability of our method, we evaluate predictive uncertainty on ImageNet (and are the first to do so, to the best of our knowledge). Most work on uncertainty in deep learning focuses on Bayesian deep learning; we hope that the simplicity and strong empirical performance of our approach will spark more interest in non-Bayesian approaches for predictive uncertainty estimation.

2 Deep Ensembles: A Simple Recipe For Predictive Uncertainty Estimation

2.1 Problem setup and High-level summary

We assume that the training dataset $\mathcal{D}$ consists of N i.i.d. data points $\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^N$, where $\mathbf{x} \in \mathbb{R}^D$ represents the D-dimensional features. For classification problems, the label is assumed to be one of K classes, that is $y \in \{1, \ldots, K\}$. For regression problems, the label is assumed to be real-valued, that is $y \in \mathbb{R}$. Given the input features $\mathbf{x}$, we use a neural network to model the probabilistic predictive distribution $p_\theta(y|\mathbf{x})$ over the labels, where θ are the parameters of the NN.

We suggest a simple recipe: (1) use a proper scoring rule as the training criterion, (2) use *adversarial training* [18] to smooth the predictive distributions, and (3) train an *ensemble*. Let M denote the number of NNs in the ensemble and $\{\theta_m\}_{m=1}^M$ denote the parameters of the ensemble. We first describe how to train a single neural net and then explain how to train an ensemble of NNs.

2.2 Proper scoring rules

Scoring rules measure the quality of predictive uncertainty (see [17] for a review). A scoring rule assigns a numerical score to a predictive distribution $p_\theta(y|\mathbf{x})$, rewarding better calibrated predictions over worse. We shall consider scoring rules where a higher numerical score is better. Let a scoring rule be a function $S(p_\theta, (y, \mathbf{x}))$ that evaluates the quality of the predictive distribution $p_\theta(y|\mathbf{x})$ relative to an event $y|\mathbf{x} \sim q(y|\mathbf{x})$ where $q(y, \mathbf{x})$ denotes the true distribution on $(y, \mathbf{x})$-tuples. The expected scoring rule is then $S(p_\theta, q) = \int q(y, \mathbf{x}) S(p_\theta, (y, \mathbf{x})) dy d\mathbf{x}$. A *proper scoring rule* is one where $S(p_\theta, q) \leq S(q, q)$ with equality if and only if $p_\theta(y|\mathbf{x}) = q(y|\mathbf{x})$, for all p_θ and q. NNs can then be trained according to measure that encourages calibration of predictive uncertainty by minimizing the loss $\mathcal{L}(\theta) = -S(p_\theta, q)$.

It turns out many common NN loss functions are proper scoring rules. For example, when maximizing likelihood, the score function is $S(p_\theta, (y, \mathbf{x})) = \log p_\theta(y|\mathbf{x})$, and this is a proper scoring rule due to Gibbs inequality: $S(p_\theta, q) = \mathbb{E}_{q(\mathbf{x})} q(y|\mathbf{x}) \log p_\theta(y|\mathbf{x}) \leq \mathbb{E}_{q(\mathbf{x})} q(y|\mathbf{x}) \log q(y|\mathbf{x})$. In the case of multi-class K-way classification, the popular softmax cross entropy loss is equivalent to the log likelihood and is a proper scoring rule. Interestingly, $\mathcal{L}(\theta) = -S(p_\theta, (y, \mathbf{x})) = K^{-1} \sum_{k=1}^K \big(\delta_{k=y} - p_\theta(y = k|\mathbf{x})\big)^2$, i.e., minimizing the squared error between the predictive probability of a label and one-hot encoding of the correct label, is also a proper scoring rule known as the Brier score [9]. This provides justification for this common trick for training NNs by minimizing the squared error between a binary label and its associated probability and shows it is, in fact, a well defined loss with desirable properties.[1]

2.2.1 Training criterion for regression

For regression problems, NNs usually output a single value say $\mu(\mathbf{x})$ and the parameters are optimized to minimize the mean squared error (MSE) on the training set, given by $\sum_{n=1}^N \big(y_n - \mu(\mathbf{x}_n)\big)^2$. However, the MSE does not capture predictive uncertainty. Following [47], we use a network that outputs two values in the final layer, corresponding to the predicted mean $\mu(\mathbf{x})$ and variance[2] $\sigma^2(\mathbf{x}) > 0$. By treating the observed value as a sample from a (heteroscedastic) Gaussian distribution with the predicted mean and variance, we minimize the negative log-likelihood criterion:

$$-\log p_\theta(y_n|\mathbf{x}_n) = \frac{\log \sigma_\theta^2(\mathbf{x})}{2} + \frac{\big(y - \mu_\theta(\mathbf{x})\big)^2}{2\sigma_\theta^2(\mathbf{x})} + \text{constant}. \tag{1}$$

We found the above to perform satisfactorily in our experiments. However, two simple extensions are worth further investigation: (i) Maximum likelihood estimation over $\mu_\theta(\mathbf{x})$ and $\sigma_\theta^2(\mathbf{x})$ might overfit; one could impose a prior and perform maximum-a-posteriori (MAP) estimation. (ii) In cases where the Gaussian is too-restrictive, one could use a complex distribution e.g. mixture density network [5] or a heavy-tailed distribution.

[1] Indeed as noted in Gneiting and Raftery [17], it can be shown that asymptotically maximizing any proper scoring rule recovers true parameter values.

[2] We enforce the positivity constraint on the variance by passing the second output through the *softplus* function $\log(1 + \exp(\cdot))$, and add a minimum variance (e.g. 10^{-6}) for numerical stability.